

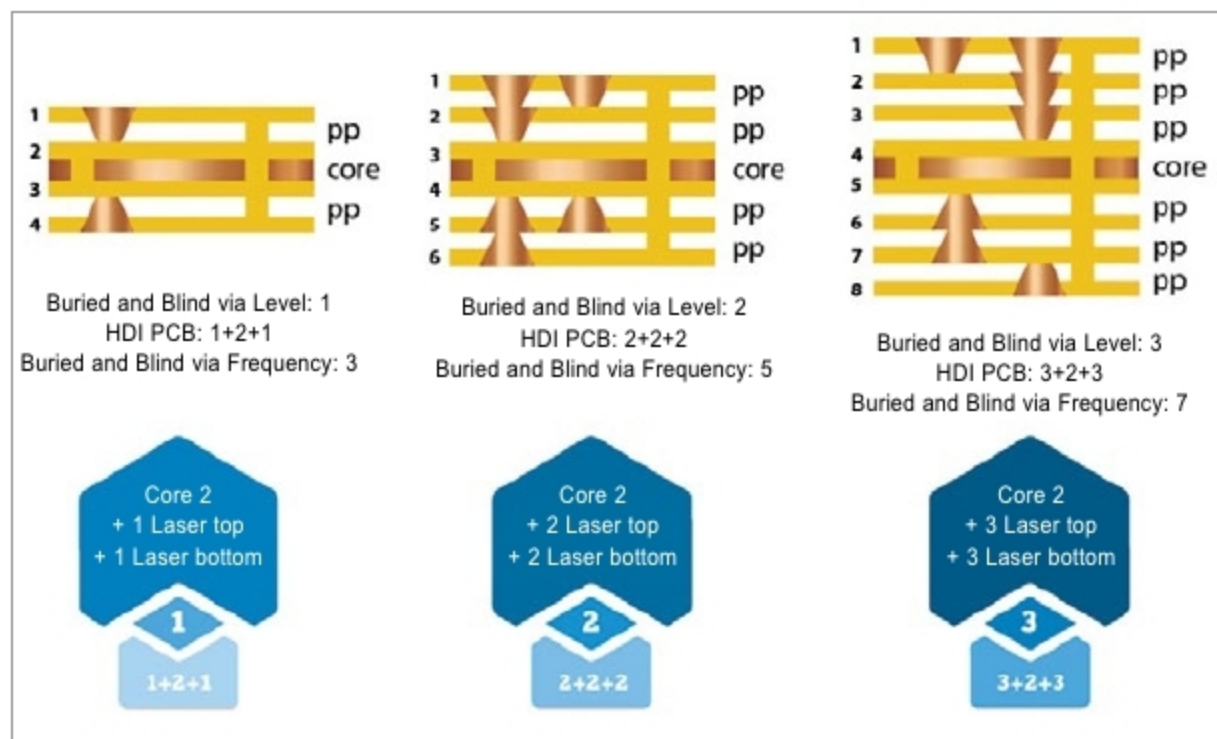


Fig. 2: HDI PCB with blind vias as stack vias and stagger vias

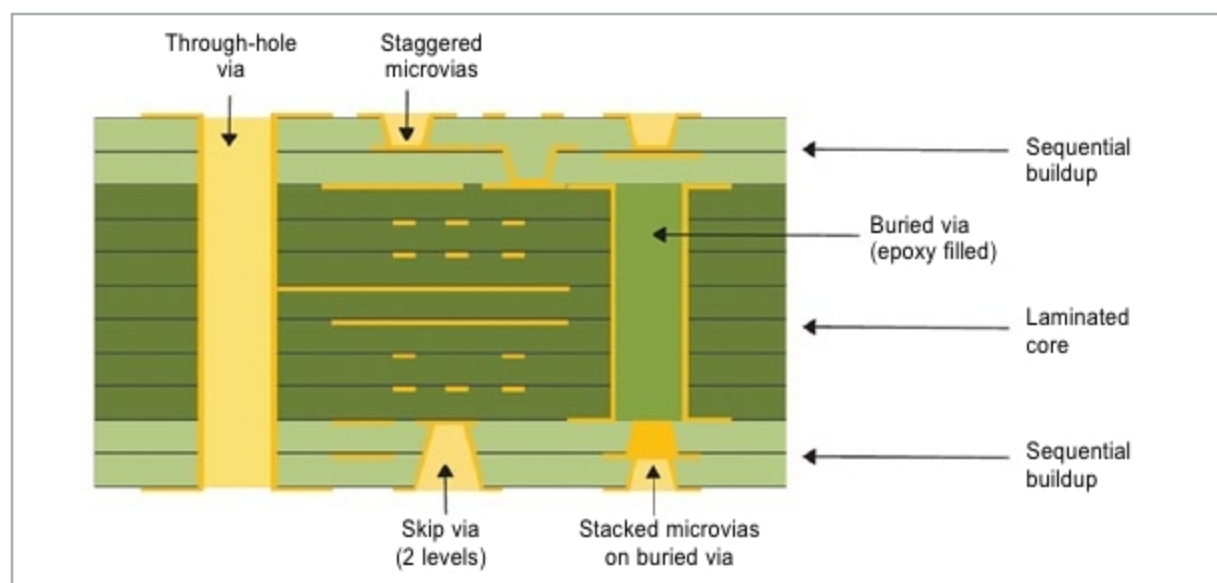


Fig. 3: HDI PCB drawing

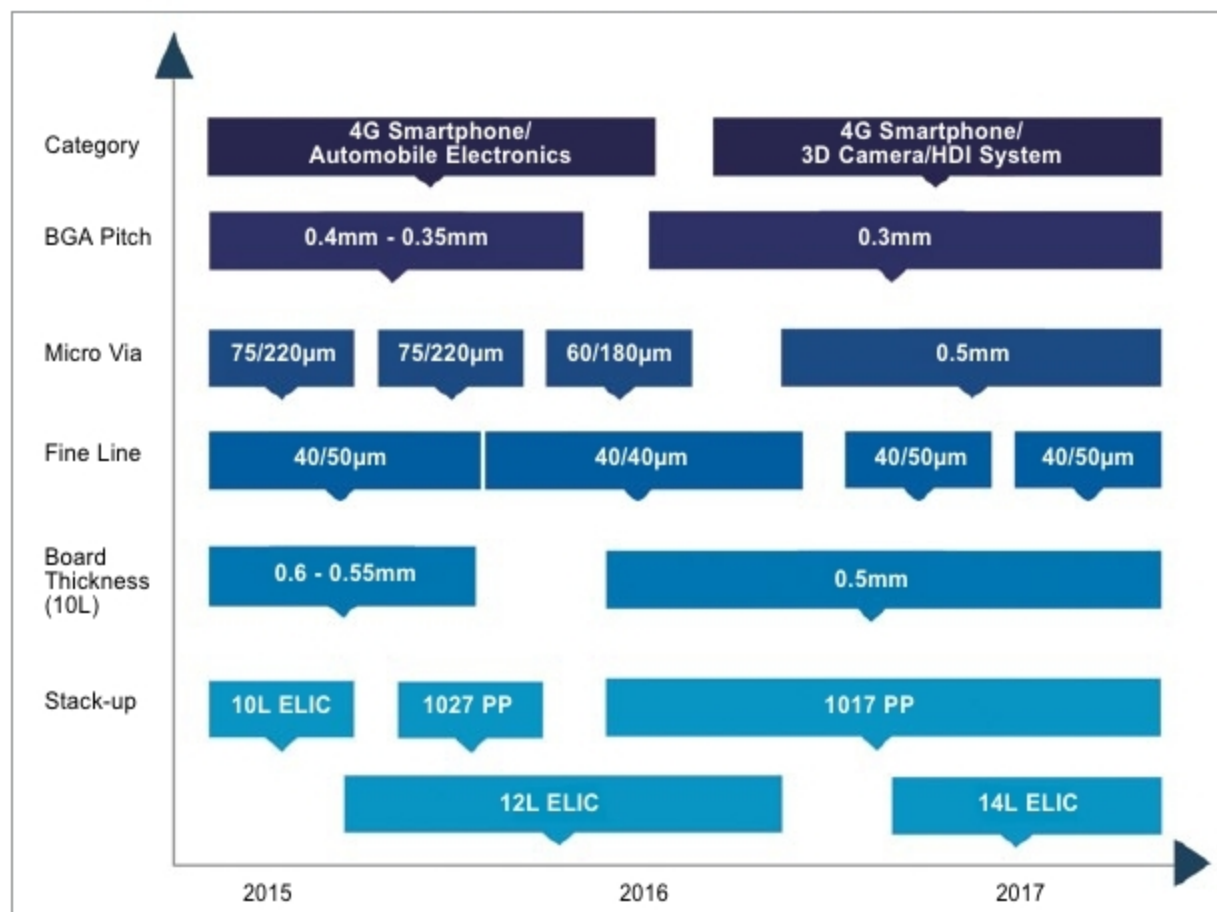


Fig. 4: HDI PCB trends and applications

TABLE I
DIFFERENT
HDI STRUCTURES

Structure	Details
1+N+1	Contains one buildup of high-density interconnection layers.
i+N+i (i ≥ 2)	Contains two or more buildups of high-density interconnection layers. Micro-vias on different layers can be staggered or stacked. Copper-filled stacked micro-via structures are commonly seen in challenging designs.
Any layer HDI	All layers of a PCB are high-density interconnection layers which allows the conductors on any layer of the PCB to be interconnected freely with copper-filled stacked micro-via structures (any layer via). This provides a reliable interconnect solution for highly-complex large pin-count devices, such as CPU and GPU chips utilised on handheld and mobile devices.

HDI PCB trends and applications

The graph in Fig. 4 shows the trends of HDI development and its main applications.

HDI leads to:

- Line reduction
- Space reduction
- BGA pitch reduction
- Via diameter reduction
- Pre-preg thickness reduction
- Board thickness reduction
- Based copper reduction

Limitations of HDI PCB are directly linked to plated holes reliability. Limitation of blind via is the aspect ratio. Aspect ratio is the division of material thickness by hole PTH diameter.

The main problem is the plating process. To assume good hole plating integrity, aspect ratio is limited to 1:0.8 for blind vias. Whereas, the advanced value is 1:1.

Due to this aspect ratio limitation, PCB manufacturers select special pre-pregs (very thin, with appropriate reaction to laser drilling). HDI multi-layers must be built up with appropriate materials. The standard PP used for standard ML like 7628, 1080 and 2116 is too thick for micro-vias.

Standard pre-pregs contain fibre-glass, which is huge for laser drill-